Identifying the Effects of Funded vs. Unfunded Fiscal Shocks

Incomplete and very preliminary!

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Resurgence of interest in fiscal explanation of inflation

- Inflation and large fiscal deficits post-COVID
- Fiscal Theory of Price Level
  - Cochrane (2023), Barro and Bianchi (2024)

Distinction between funded and unfunded deficits

- **Funded deficits:**
    - current deficit is offset by present value of future surpluses
    - lump-sum taxes, no frictions: Ricardian equivalence (deficits are neutral)
    - monetary policy is active, fiscal policy passive
- **Unfunded deficit shock:**
    - current deficit is not fully offset by future surpluses
    - inflationary (FTPL), expansionary (boost aggregate demand)
    - fiscal policy is active, monetary policy is passive

Research question

- Can we identify funded vs unfunded fiscal deficit shocks?
- What are the effects of these shocks on economic activity?

Existing approaches

DSGE approach: Bianchi, Faccini & Melosi (2023), Smets & Wouters (2024), Balke & Zarazaga (2024)

- Impose strong restrictions on the data generating process
- Many are auxiliary assumptions about the stochastic processes of shocks for which theory provides no guidance

What we do

- Minimal restrictions on the data generating process
    - use VAR to model dynamics of the data
- Minimal assumptions used to identify funded vs unfunded fiscal shocks
    - mixture of zero and sign restrictions
    - Keep the framework as **agnostic** as possible

Inflation and VARs

- **Conventional monetary VARs**
  - Unexpected inflation is typically attributed to monetary or other structural shocks.
  - Traditional VARs often omit fiscal variables, presume fiscal policy is neutral.
  - Shocks to primary surpluses are typically not treated as independent drivers of inflation.
- **Fiscal VARs**
  - Typically, do not separately identify funded vs unfunded shocks
  - Indirectly identified funded vs unfunded shocks when using regime switching VARs or time varying parameter VARs

Our contribution

1. **FTPL / policy-regime literature:**
   by developing a data-driven identification of funded vs unfunded fiscal shocks.
   (Leeper (1991); Sims (1994); Woodford (1995); Cochrane (2001, 2011))
2. **Empirical fiscal SVAR:**
   assess the contribution of deficit shocks to output, inflation, and interest rates
   (Blanchard & Perotti (2002); Ramey (2011); Caldara & Kamps (2017))
3. **DSGE-based identification:**
   by proposing a parsimonious structural VAR as an empirical alternative to large DSGEs.
   (Smets & Wouters (2024); Balke & Zarazaga (2024); Bianchi, Faccini & Melosi (2022))

Our findings

1. Funded deficit shocks "look" neutral
2. Unfunded deficit shocks on impact increase inflation, real GDP growth, and lower nominal returns on government debt
3. Fiscal shocks are only moderate contributor to historical fluctuations in inflation
4. Most of the fluctuations of fiscal variables are "endogenous" (i.e. government surpluses and debt respond to other shocks)
5. Results are fairly robust to alternative ways of implementing identification

Government Budget Equation

Flow budget equation

$$S_t + V_t \;=\; \frac{R^V_t}{\Pi_t\, G_t}\, V_{t-1}$$

Notation

S_t: real primary surplus over GDP
V_t: real market value of debt over GDP
R^V_t: gross nominal return on government debt
Π_t: gross inflation
G_t: real GDP growth factor

Government debt and future surpluses

Implies value of debt is equal to PV of expected future surpluses

$$V_t \ = \ E_t \sum_{j=1}^{\infty} (\prod_{i=1}^{j} \frac{R^V_{t+i}}{\Pi_{t+i} \, G_{t+i}})^{-1} \, S_{t+j}$$

V_t: present value of future primary surpluses

$S_t + V_t$: PV of current and future primary surpluses

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V_t: present value of future primary surpluses
$S_t + V_t$: PV of current and future primary surpluses

Definition

Funded deficit shock	**Unfunded deficit shock**
$\Delta E_t (S_t + V_t) = 0$	$\Delta E_t (S_t + V_t) < 0$
Fiscally neutral	Fiscally non-neutral
$V_t \uparrow$ one-for-one with deficit shock	No one-for-one change in V_t with deficit shock

Transmission of Fiscal deficit shocks to Economic Activity

Government Flow Budget Equation

$$S_t + V_t = \frac{R_t^V}{\Pi_t \, G_t} V_{t-1}$$

Funded deficit shock

- $\Delta E_t(S_t + V_t) = 0$
- R_t^V,Π_t,G_t individually may change, but $\frac{R_t^V}{\Pi_t G_t}$ unchanged
- No frictions: Ricardian equivalence, funded deficit shocks are neutral (no effect on R_t^V,Π_t, or G_t)
- Real effects due to myopia, liquidity constraints, distortionary taxes, distributional consequences

Transmission of Fiscal deficit shocks to Economic Activity

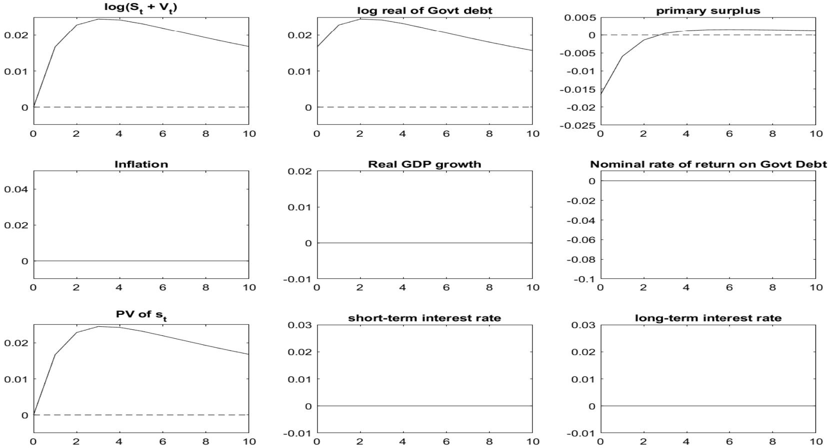
Government Flow Budget Equation

$$S_t + V_t \; = \; \frac{R_t^V}{\Pi_t \, G_t} \, V_{t-1}$$

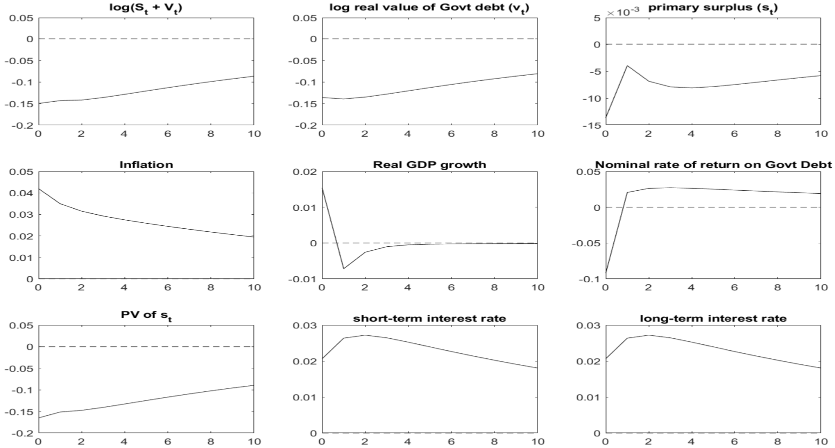
Unfunded deficit shock

- $S_t + V_t \downarrow \implies \frac{R_t^V}{\Pi_t G_t} \downarrow$
- No frictions: FTPL $\implies \Pi_t \uparrow$
- Real effects due to frictions (price stickiness, myopia, liquidity constraints, distortionary taxes, etc).

Example: New Keynesian Model Response to a Funded Deficit Shock



Example: New Keynesian Model Response to an Unfunded Deficit Shock



Data & SVAR

Sample. U.S., annual, 1959–2023; SVAR with 2 lags.

Variables in the SVAR

- $\log(S_t + V_t)$
- Primary surplus *(constructed)*, $s_t = \log(S_t + V_t) - \rho \log(V_t)$
  - $\rho \approx (\frac{R^V}{\Pi G})^{-1}$, set $\rho = 0.9999$
- Inflation ($\log(\Pi_t)$)
- Real GDP growth ($\log(G_t)$)
- 3-month T-bill rate
- 10-year Treasury yield

Data & SVAR

Companion form of the VAR

$$Y_t = A\,Y_{t-1} + \epsilon_t$$

$E(\epsilon_t \epsilon_t') = \Sigma$

Structural shocks

$$\epsilon_t = A_0 u_t$$

- u_t orthogonal ($E[u_t u_t'] = I$)
- pick A_0 to satisfy a combination of zero and sign restrictions ($A_0 A_0' = \Sigma$)

Identification Strategy: Least restrictive

Funded shock

$$\boxed{\Delta E_t \log(S_t + V_t)\big|_{\text{funded deficit shock, horizon=0}} = 0}$$

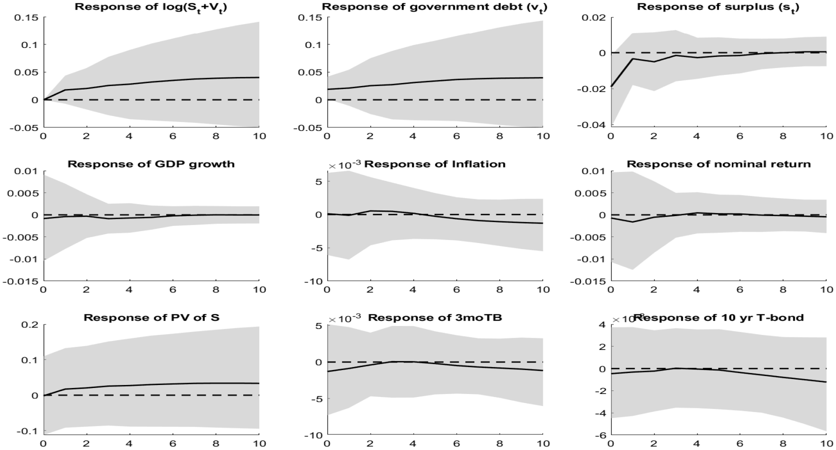
Unfunded shock

$$\boxed{\Delta E_t \log(V_t)\big|_{\text{unfunded deficit shock, horizon=0}} < 0, \quad \Delta E_t s_t\big|_{\text{unfunded deficit shock, horizon=0}} < 0}$$

This implies that unfunded deficit shock affects current surplus and PV of future surpluses in same direction on impact:

$$\Delta E_t \log(S_t + V_t)\big|_{\text{unfunded surplus shock, horizon=0}} < 0$$

Empirical IRFs: Funded Shock (90% confidence interval)



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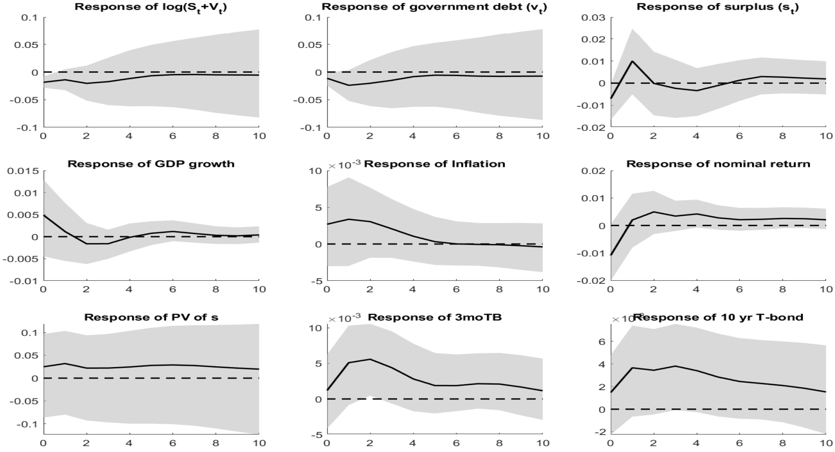
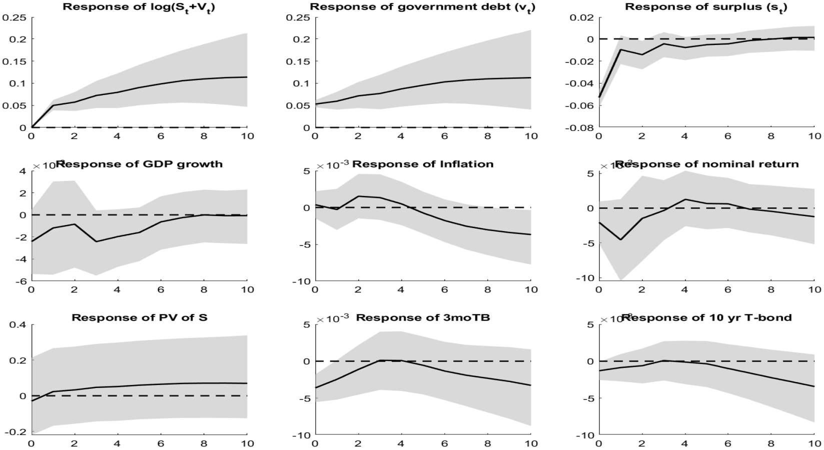


Empirical IRFs: Choleski (funded shock-deficits first)



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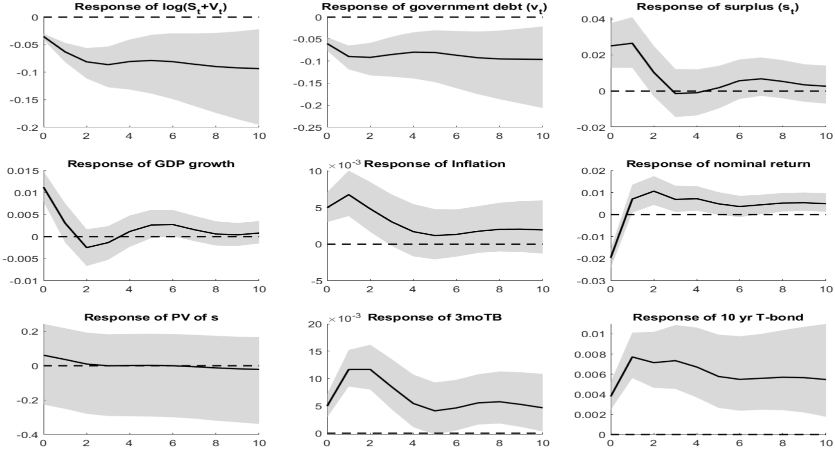
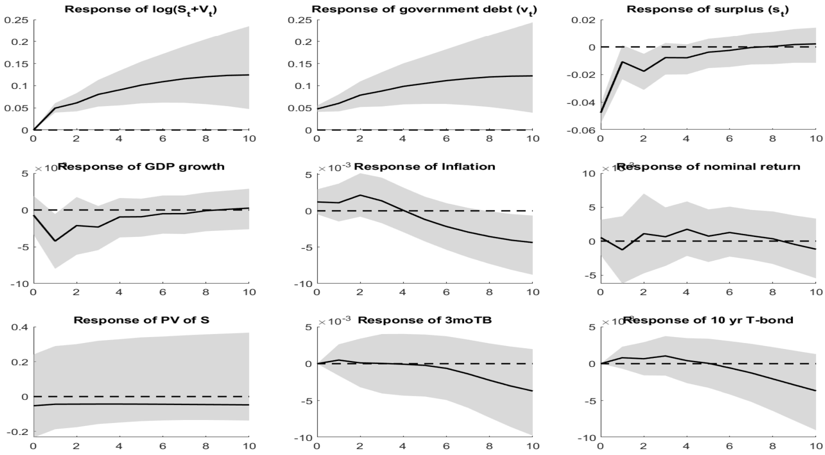
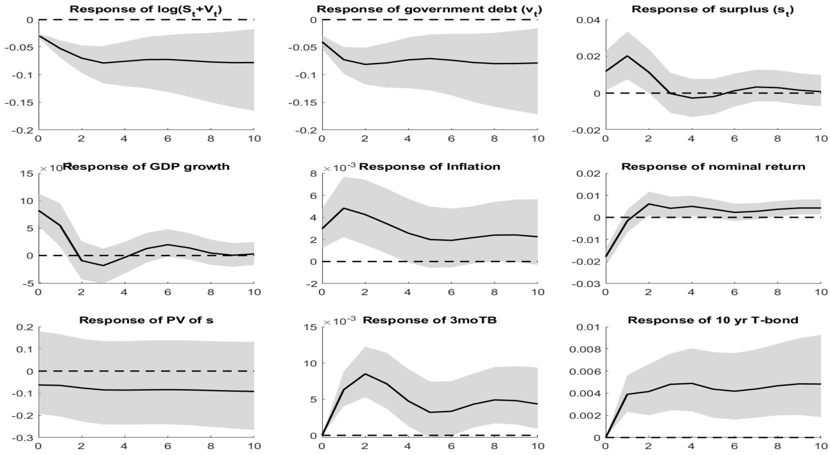


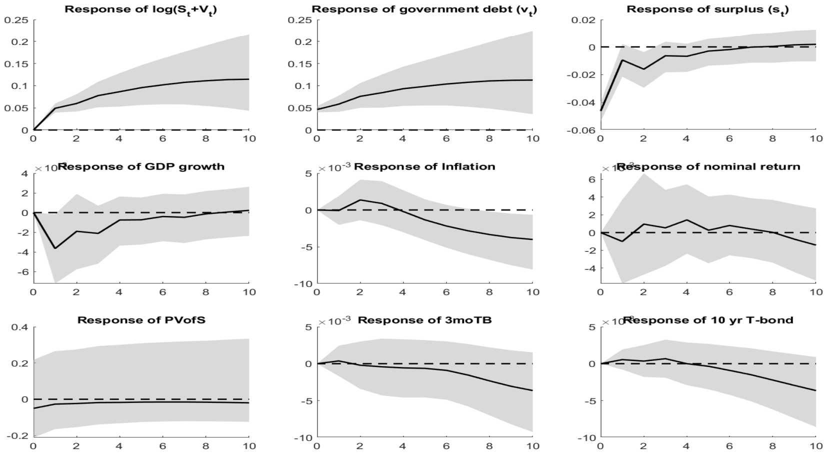
Empirical IRFs: Choleski (funded shock-deficits middle)



Empirical IRFs: Choleski (unfunded shock-deficits middle)



Empirical IRFs: Choleski (funded shock-deficits last)



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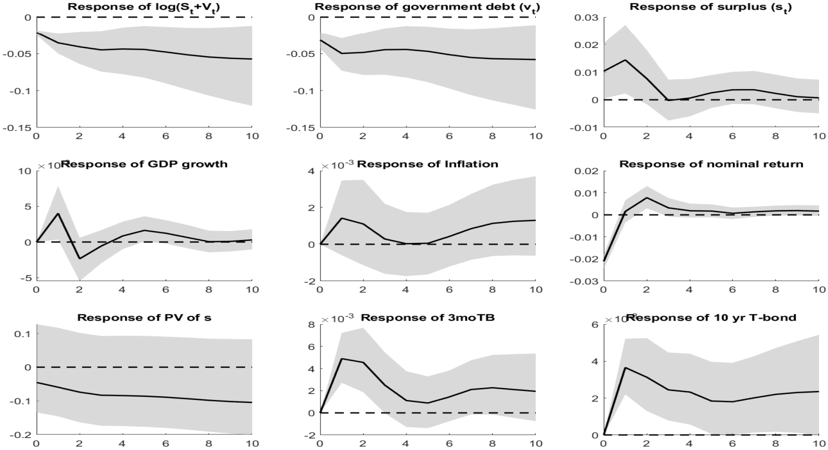
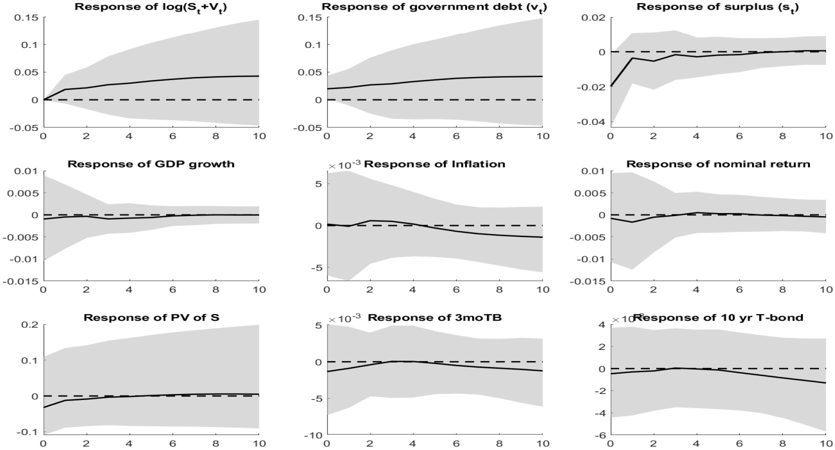
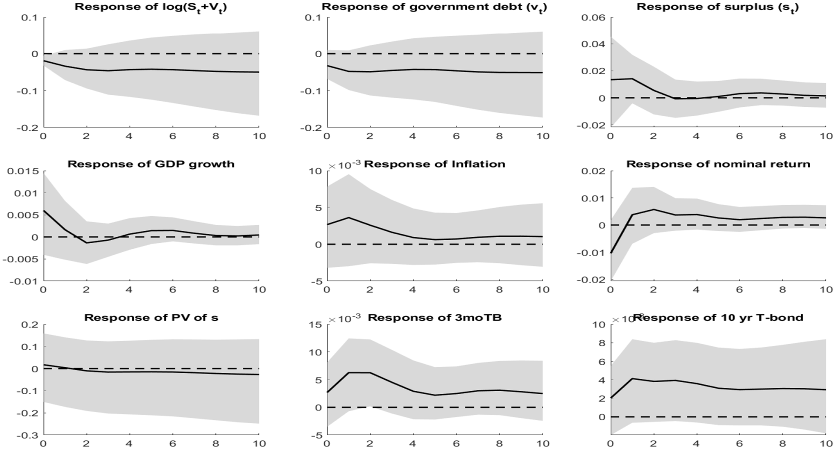


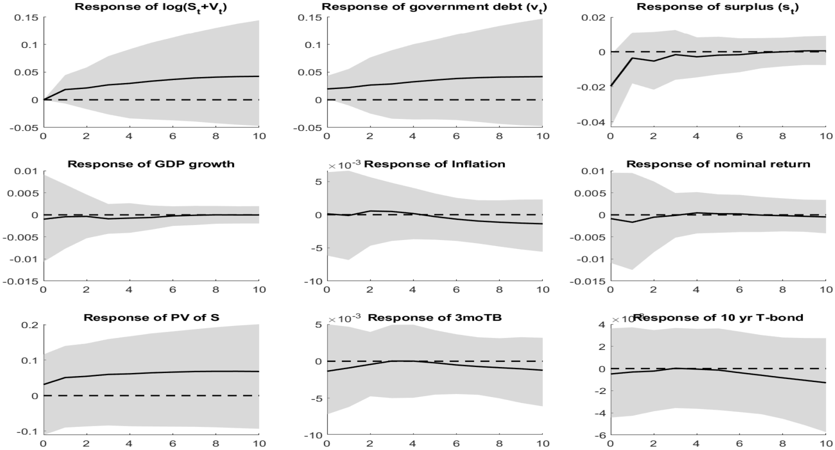
Empirical IRFs: minimum restriction (funded shock



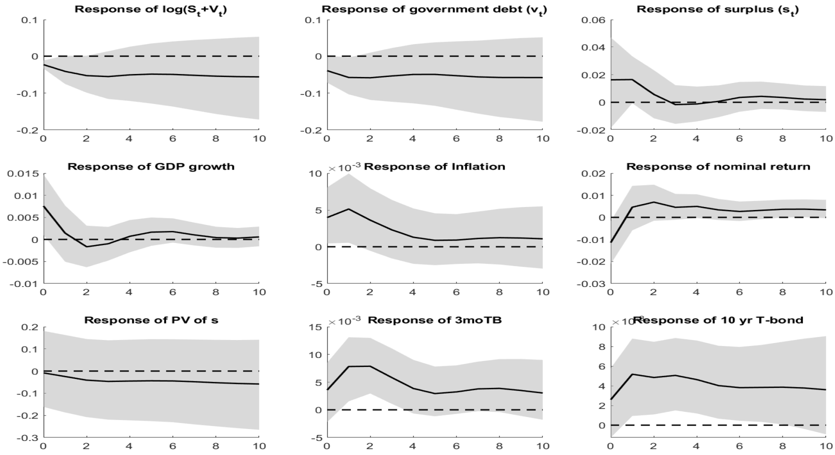
Empirical IRFs: minimum restriction(unfunded shock)



Empirical IRFs: sign restriction (funded shock



Empirical IRFs: sign restriction(unfunded shock)

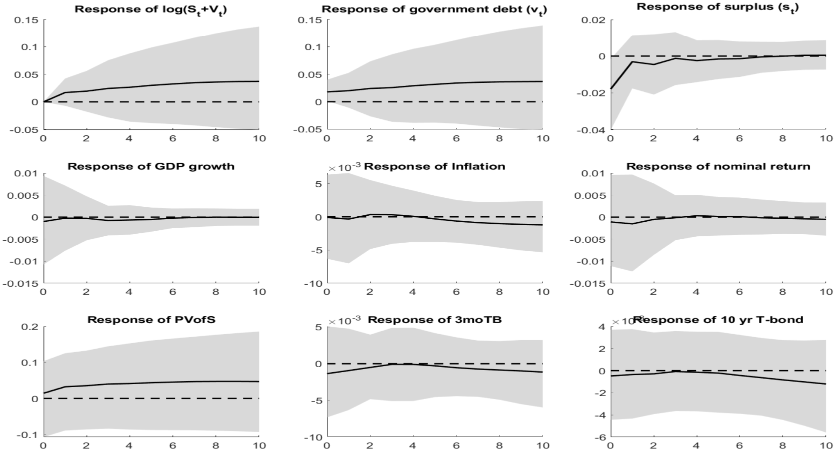


Identification Strategy: additional restrictions

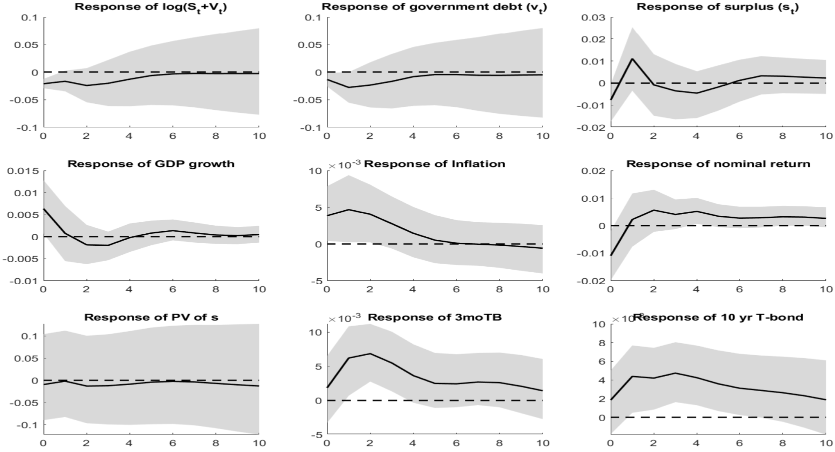
- Use the predictions from simple New Keynesian model
- Additional sign restrictions for unfunded deficit shock:

$$\Delta E_t \log(\Pi_t) > 0,\ \Delta E_t \log(G_t) > 0,\ \Delta E_t \log(R_t^V) < 0$$

Empirical IRFs with additional sign restrictions: Funded Shock



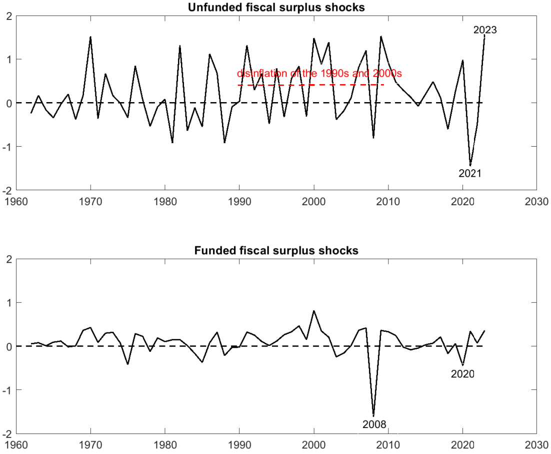
Empirical IRFs with additional sign restrictions: Unfunded Shock



Summary of IRF results for benchmark specification

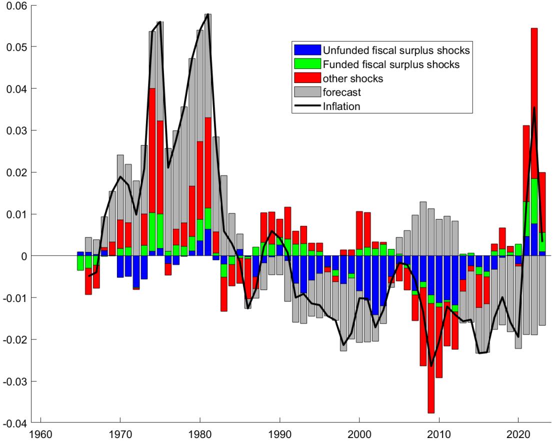
- Funded deficit shocks look pretty "Ricardian"–not much of an effect on economic activity
- Unfunded deficit shocks appear to have stimulative effect on economic activity
  - Real GDP growth is positive on impact
  - Persistent increase in inflation
  - Relatively large decrease in nominal return on govt debt on impact
  - Persistent increase in nominal interest rates (humped shaped)

Identified unfunded and funded fiscal deficit shocks



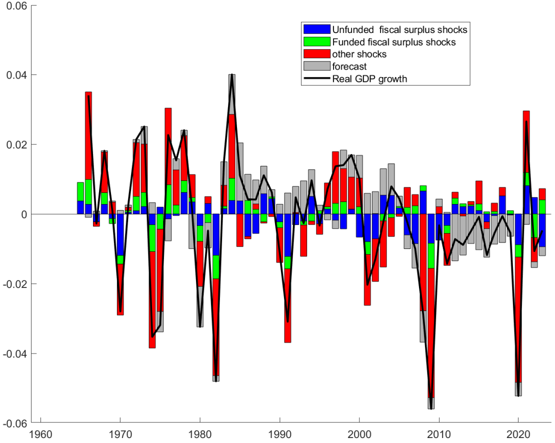
Historical Decomposition (4 year forecast error) Inflation

Blue:unfunded, Green:funded, Red:other, Gray:forecast,



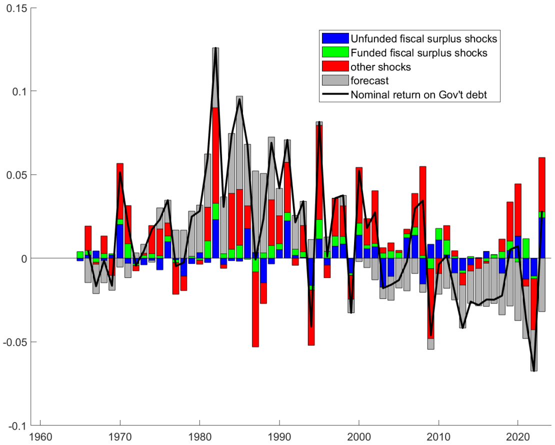
Historical Decomposition (4 year forecast error) real GDP growth

Blue:unfunded, Green:funded, Red:other, Gray:forecast,

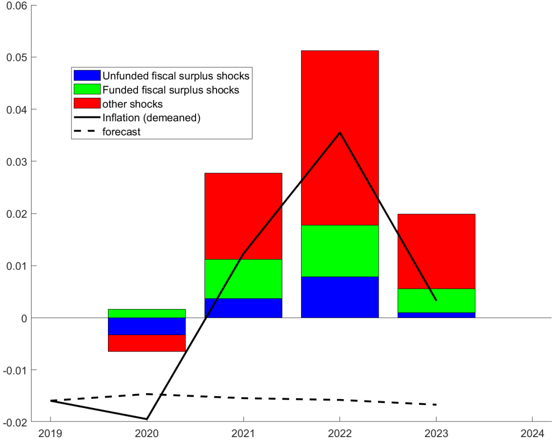


Historical Decomposition (4 year forecast error) nominal return on Govt debt

Blue:unfunded, Green:funded, Red:other, Gray:forecast,



Historical Decomposition: Inflation (COVID episode)



Variance decompositions

Table: Contribution of funded deficit shocks

horizon	$log(S_t + V_t)$	surplus (s_t)	inflation	real GDP growth	nom. return govt debt
0	0.0	13.5	15.8	13.0	5.8
1	6.3	12.0	14.6	14.2	8.4
5	10.1	13.0	14.7	14.3	5.2
10	11.5	13.0	14.7	14.1	5.1
20	11.5	13.0	14.3	13.9	3.6

Variance decompositions

Table: Contribution of unfunded deficit shocks

horizon	$log(S_t + V_t)$	surplus (s_t)	inflation	real GDP growth	nom. return govt debt
0	37.1	2.5	20.5	17.4	4.1
1	10.8	6.1	21.5	16.1	4.1
5	6.7	8.0	22.3	16.5	9.3
10	6.1	8.1	18.9	16.6	12.0
20	7.3	9.6	15.4	16.2	7.8

Summary of decomposition results

- Exogenous fiscal shocks contribute relatively small percent of the variability of economic variables
- Most of variation of fiscal policy ($log(S_t + V_t)$ and s_t) is endogenous
  - Policy "rule" more important than "shocks" to policy

Identification Strategy: Ricardian fiscal policy

Funded deficit shock: Future surpluses do all the work to offset current deficit shock

$$\Delta E_t \sum_{j=0}^{\infty} \rho^j \, s_{t+j} = 0$$

PV of current and future primary surpluses (discounted at constant rate)
is unaffected by funded deficit.

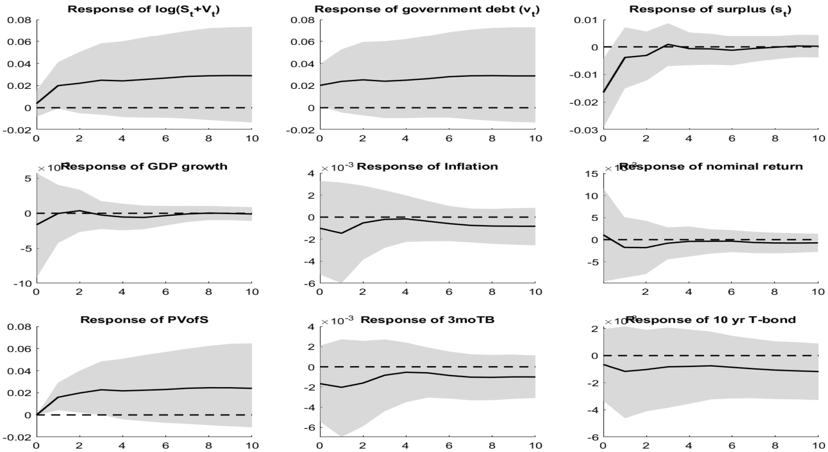
Identification Strategy: Ricardian surpluses

Funded deficit shock: Pick A_0 matrix so that

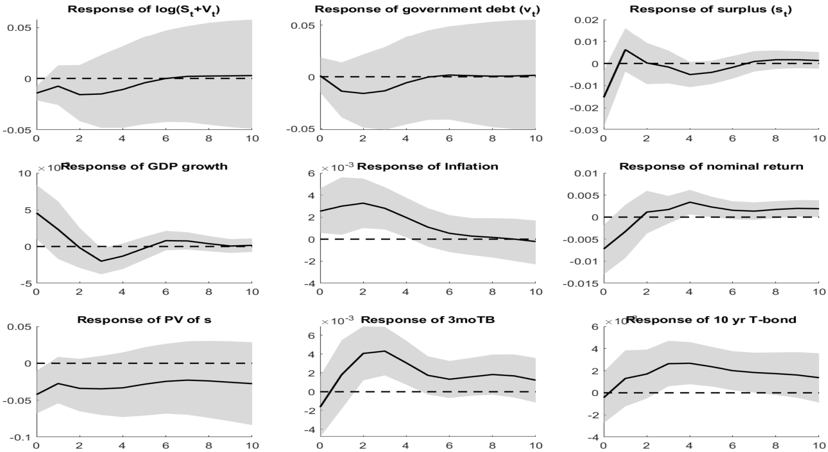
$$H^{\textit{funded surplus}} \left(I - \rho A\right)^{-1} A_0 J^{\textit{funded deficit}} u_t = 0$$

- $H^{\textit{funded surplus}}$ selects surplus (s_t) from the variables in VAR
- $J^{\textit{funded deficit}}\, u_t$ selects funded deficit shock from vector of exogenous shocks
- A_0 must also satisfy the sign restrictions for unfunded deficit shock

Empirical IRFs Ricardian surpluses: Funded Shock



Empirical IRFs Ricardian surpluses: Unfunded Shock



Ricardian surpluses

- Historical decompositions similar to benchmark specification
- Variance decompositions of similar magnitude to benchmark specification

Summary

- Estimate SVAR that identifies funded vs unfunded fiscal deficit shocks
- Funded deficit shocks look neutral – do not affect economic activity
- Unfunded deficit shocks have real affects consistent with FTPL
- Fiscal shocks only moderate contributor to historical fluctuations in inflation
- Most of the fluctuations of fiscal variables are "endogenous" (i.e. government surpluses and debt respond to other shocks)

Future work

- Quarterly data
- Add narrative approach to identification